Test 2 Practice Problems

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Questions

Test 2 will be on Monday March 2. These problems will prepare you for the test. You will have a formula sheet, which you can see on CourseDen.

Question 1

0/1 pt 99 98

Let $\vec{r}(t) = \langle t^3 + 5, 4e^{3t}, 4\sin(3t) \rangle$

Find the unit tangent vector $\vec{T}(t)$ at the point $t = 0$

$\vec{T}(0)$ = < , , >

$\vec{T}(t) =< 0, 0.70710678118655, 0.70710678118655 >$

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Question 2

0/1 pt 99 98

Find the unit tangent vector to the curve defined by

$\vec{r}(t) = \langle 2\cos(t), 2\sin(t), 5\sin^2(t) \rangle$

at $t = \pi$.

$\vec{T}(\pi) =$ ⟨0, -1, 0⟩

Use the unit tangent vector to write the parametric equations of a tangent line to the curve at $t = \pi$.

$x(t) =$ -2

$y(t) =$ $-t$

$z(t) =$ 0

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Question 3

0/1 pt 99 98

For the curve defined by

$\vec{r}(t) = \langle e^{-t}\cos(t), e^{t}\sin(t)\rangle$

find the unit tangent vector, unit normal vector, normal acceleration, and tangential acceleration at $t = \frac{\pi}{4}$.

$\vec{T}\left(\frac{\pi}{4}\right) =$ $\langle -0.20352845542175, 0.97906903118914\rangle$

$\vec{N}\left(\frac{\pi}{4}\right) =$ $\langle 0.97906903118914, 0.20352845542175\rangle$

$a_T =$ 2.9056095149346

$a_N =$ 1.2625954464331

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Question 4

0/1 pt 99 98

Find the length of the curve $\vec{r}(t) = \langle 5\cos(4t), 5\sin(4t), 3t\rangle$ for $-2 \le t \le 1$

Give your answer to two decimal places

$\int_{-2}^{1} |\vec{r}'(t)|\, dt = \int_{-2}^{1} 20.223748416157 dt = 60.67124524847$

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Question 5

0/1 pt 99 98

Find the length of the curve $\bar{r}(t) = \langle 4\ln(t), t^2, 4t\rangle$ for $1 \le t \le 3$

12.394449154672

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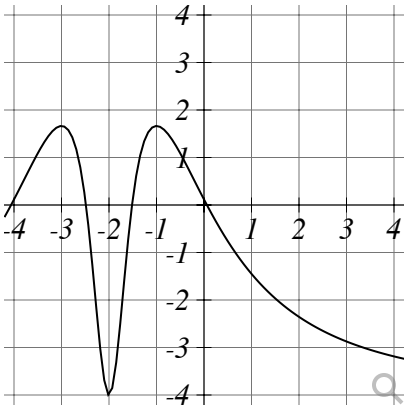
Question 6

0/1 pt 99 98

For the function graphed below, estimate the point of greatest curvature



$x =$ _____ -2

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Question 7

0/1 pt 99 98

Find the curvature of the curve $\vec{r}(t) = \langle -2t, 4t^2, 3t^3 \rangle$ at the point $t = 2$.

Give your answer to 2 decimal places.

$$\kappa = \frac{|r'(2) \times r''(2)|}{|r'(2)|^3} = \frac{|< -2, 16, 36 > \times < 0, 8, 36 >|}{(39.446165846632)^3} = \frac{|< 288, 72, -16 >|}{(39.446165846632)^3} = \frac{297.29446681699}{(39.446165846632)^3} = 0.0048436464714701$$

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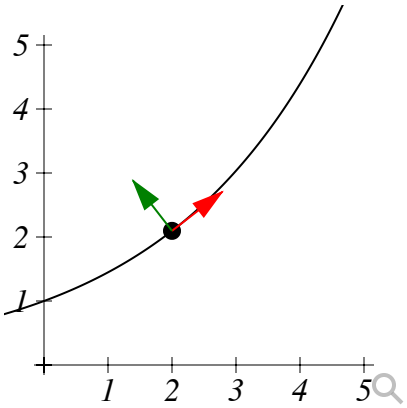
Question 8

0/1 pt 99 98



In this question you may round each entry to the nearest hundredth where necessary.

In parts (b) and (c), note that we can measure the curvature of a curve with $k(t) = \dfrac{\left|\vec{T}'(t)\right|}{\left|\vec{r}'(t)\right|}$.

The curve shown above in the xy plane is given by $\vec{r}(t) =< t, e^{0.37t} >$

a) Compute $\vec{T}(2)$ and $\vec{N}(2)$.

$\vec{T}(2) =$ _____

$\langle 0.79022518304873, 0.61281657947187 \rangle$ or $(0.79022518304873, 0.61281657947187)$

$\vec{N}(2) =$ _____

$\langle -0.61281657947187, 0.79022518304873 \rangle$ or $(-0.61281657947187, 0.79022518304873)$

b) What is the curvature of the above curve when $t = 2$?

$k(2) =$ _____ 0.14159044998587

Submit Question

Question 9

0/1 pt 99 98

Given the vector function $\vec{r}(t) = \langle -5t, t^4, -3t^3 + 5 \rangle$

Find the velocity and acceleration vectors at $t = 1$

$\vec{v}(1)$ =

$\left[\quad \quad \quad \right]$

$[-5 \quad 4 \quad -9]$

$\vec{a}(1)$ =

$\left[\quad \quad \quad \right]$

$[0 \quad 12 \quad -18]$

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Question 10

0/1 pt 99 98

Find the tangential and normal components of the acceleration vector for the curve $\vec{r}(t) = \langle -4t, t^5, -3t^3 \rangle$ at the point $t = 1$

$\vec{a}(1) =$ 23.72036546314 $\vec{T}$ + 12.702136131182 $\vec{N}$

Give your answers to two decimal places

Submit Question

Question 11

0/1 pt 99 98

For the curve defined by

$\vec{r}(t) = \langle e^{-t}, 5t, e^t \rangle$

find the unit tangent vector, normal acceleration, and tangential acceleration at $t = 0$.

$\vec{T}(t) =$

$\langle -0.19245008972988, 0.96225044864938, 0.19245008972988 \rangle$

$a_T =$ 0

$a_N =$ 1.4142135623731

Submit Question

Question 12

0/1 pt 99 98

Given $f(x,y) = -4x^5 - 6xy^6 + 2y^4$, find

$f_x(x,y)$ = $-20x^4 - 6y^6$

$f_y(x,y)$ = $-36xy^5 + 8y^3$

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Question 13

0/1 pt 99 98

Given $f(x,y) = -6x^6 - x^2y^5 - 2y^3$, find

$f_x(x,y)$ = $-36x^5 - 2xy^5$

$f_y(x,y)$ = $-5x^2y^4 - 6y^2$

$f_{xx}(x,y)$ = $-180x^4 - 2y^5$

$f_{xy}(x,y)$ = $-10xy^4$

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Question 14

0/1 pt 99 98

Find the tangent plane to the equation $z = -5x^2 - y^2 - 4y$ at the point $(4, -1, -77)$

z = $-40x - 2y + 81$

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Question 15

0/1 pt 99 98

Find the tangent plane to the equation $z = 2e^{x^2-2y}$ at the point $(4, 8, 2)$

$z =$ $16x - 4y - 30$

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Question 16

0/1 pt 99 98

Use linear approximation to $x^3 + 7xy + y^3$ at the point $(5, 4, 329)$, and use it to approximate

$f(5.14, 4.22) =$ 361.68

Submit Question

Question 17

0/1 pt 99 98

Part 1 of 2

Find the linearization of the function $f(x, y) = \sqrt{87 - 3x^2 - 4y^2}$ at the point $(-5, -1)$.

- $L(x, y) =$ $\dfrac{87 + 15x + 4y}{\sqrt{8}}$

Part 2 of 2

Use the linear approximation above to estimate the value of $f(-4.9, -1.1)$.

- $f(-4.9, -1.1) =$ 3.217

Note: Your answer is judged to 3 decimal places of accuracy.

Submit Question

Question 18

0/1 pt 99 98

Find $\frac{dz}{dt}$ given:

$z = xe^{3y}, \quad x = t^5, \quad y = -1 + 2t$

$\frac{dz}{dt}$ = $5e^{3(-1+2t)}t^4 + 6t^5e^{3(-1+2t)}$

Your answer should only involve the variable *t*

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Question 19

0/1 pt 99 98

The temperature at a point on a metal plate is given by $T(x, y) = \frac{x}{x^2 + y^2}$. Use this information to answer the questions below. Write all vectors in component form and give all answers in exact form.

a. Find the direction of greatest increase of heat at the point $(-2, -1)$.

Direction of greatest increase is $\left\langle -\frac{3}{25}, -\frac{4}{25} \right\rangle$

b. Find the rate of maximum temperature increase.

Rate of maximum increase is $\frac{1}{5}$

c. Find the direction of greatest decrease of at the point $(-2, -1)$.

Direction of greatest decrease is $\left\langle --\frac{3}{25}, --\frac{4}{25} \right\rangle$

d. Find the rate of maximum decrease of temperature.

Rate of maximum decrease is $-\frac{1}{5}$

Submit All Parts

Question 20

0/1 pt 99 98

Given $f(x,y) = \sin(x+y)$ where $x = s^3t^5, y = 3s - 5t$. Find

$f_s(x(s,t), y(s,t))$ =

$\cos\left(s^3t^5 + 3s - 5t\right) \cdot \left(3s^2t^5 + 3\right)$

$f_t(x(s,t), y(s,t))$ =

$\cos\left(s^3t^5 + 3s - 5t\right) \cdot \left(5s^3t^4 - 5\right)$

Note: This question is looking for the answer to be only in terms of s and t

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